PIZZA

Hello!

In this project i have used SQL to solve queries that were related to pizza sales analysis.

Database Brief

The database structure follows a real world transactional model where:

- **Orders** store order timestamps.
- **Order_Details** record the items purchased in each order.
- **Pizzas** contain pricing and size information.
- **Pizza_Types** provide descriptive details such as pizza names, categories, and ingredients.

Retrieve the total number of orders placed.

```
SELECT
    COUNT(order_id) AS total_orders
FROM
    orders;
```

Result Grid

	total_orders
▶	21350

Calculate the total revenue generated from pizza sales.

```sql
SELECT
    ROUND(SUM(o.quantity * p.price), 2)
FROM
    order_details AS o
        JOIN
    pizzas AS p ON o.pizza_id = p.pizza_id;
```

Result Grid | Filter R

ROUND(SUM(o.quantity * p.price), 2)
817860.05

Identify the highest-priced pizza.

```
SELECT
    *
FROM
    pizzas
ORDER BY price DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export:

pizza_id	pizza_type_id	size	price
the_greek_xxl	the_greek	XXL	35.95

Identify the most common pizza size ordered.

```sql
SELECT
    p.size, COUNT(o.order_details_id) AS max_ordered_size
FROM
    pizzas AS p
        JOIN
    order_details AS o ON p.pizza_id = o.pizza_id
GROUP BY size
ORDER BY max_ordered_size DESC
LIMIT 1;
```

Result Grid | Filter Rov

size	max_ordered_size
L	18526

List the top 5 most ordered pizza types along with their quantities

```
SELECT
    p.name, SUM(o.quantity) AS max_qty
FROM
    pizza_types AS p
        JOIN
    pizzas ON p.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details AS o ON o.pizza_id = pizzas.pizza_id
GROUP BY p.name
ORDER BY max_qty DESC
LIMIT 5;
```

Result Grid | Filter Rows:

name	max_qty
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Join the necessary tables to find the total quantity of each pizza category ordered.

```sql
SELECT
    p.category, SUM(o.quantity) AS total_qty
FROM
    pizza_types AS p
        JOIN
    pizzas ON p.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details AS o ON o.pizza_id = pizzas.pizza_id
GROUP BY p.category
ORDER BY total_qty DESC;
```

Result Grid | Filter Rows:

category	total_qty
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Determine the distribution of orders by hour of the day.

```
SELECT
    HOUR(time), COUNT(order_id)
FROM
    orders
GROUP BY HOUR(time);
-- question8 --
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

Result Grid | Filter Rows:

category	COUNT(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

Join relevant tables to find the category-wise distribution of pizzas.

```sql
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

Result Grid | Filter Rows:

category	COUNT(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

Group the orders by date and calculate the average number of pizzas ordered per day.

```sql
SELECT
    ROUND(AVG(quantity), 0)
FROM
    (SELECT
        o.date, SUM(od.quantity) AS quantity
    FROM
        orders AS o
    JOIN order_details AS od ON o.order_id = od.order_id
    GROUP BY o.date) AS order_quantity;
```

Result Grid | Filter Rows:

average_pizzaorder_perday
138

Determine the top 3 most ordered pizza types based on revenue.

```sql
SELECT
    pt.name, SUM(od.quantity * p.price) AS revenue
FROM
    pizza_types AS pt
        JOIN
    pizzas AS p ON pt.pizza_type_id = p.pizza_type_id
        JOIN
    order_details AS od ON p.pizza_id = od.pizza_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT
    pt.category,
    ROUND(SUM(od.quantity * p.price) / (SELECT
                ROUND(SUM(od.quantity * p.price), 2) AS total_sales
            FROM
                order_details AS od
                    JOIN
                pizzas AS p ON p.pizza_id = od.pizza_id) * 100,
        2) AS revenue
FROM
    pizza_types AS pt
        JOIN
    pizzas AS p ON pt.pizza_type_id = p.pizza_type_id
        JOIN
    order_details AS od ON p.pizza_id = od.pizza_id
GROUP BY pt.category
ORDER BY revenue DESC;
```

Result Grid | Filter Rows:

category	revenue
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

Analyze the cumulative revenue generated over time.

```sql
select date,sum(revenue)
over (order by date) as cumulative_revenue
from
(select o.date,sum(od.quantity*p.price) as revenue
from
order_details as od join pizzas as p
on od.pizza_id=p.pizza_id
join
orders as o
on o.order_id=od.order_id
group by o.date) as sales;
```

Result Grid | Filter Rows:

date	cumulative_revenue
2015-01-01	2713.8500000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5

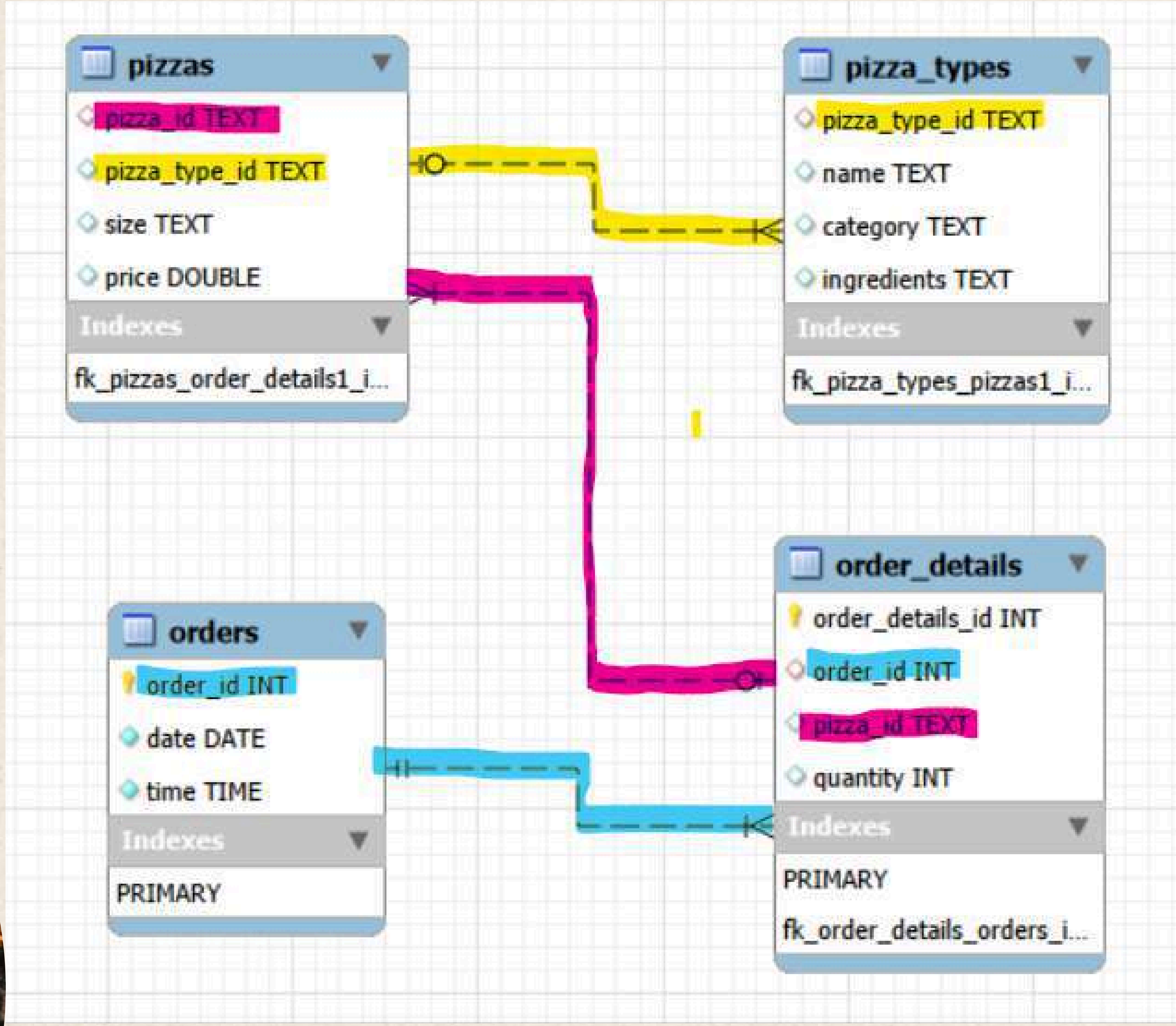
Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
select name,revenue
from
(select category,name,revenue,rank()
over(partition by category order by revenue desc) as rn
from
(select pt.category,pt.name,sum((od.quantity)*p.price) as revenue
from
pizza_types as pt
join
pizzas as p
on pt.pizza_type_id=p.pizza_type_id
join
 order_details as od
on od.pizza_id=p.pizza_id
group by pt.category,pt.name)as a) as b
where rn<=3;
```

Result Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25
The Pepperoni Pizza	30161.75
The Spicy Italian Pizza	34831.25
The Italian Supreme Pizza	33476.75
The Sicilian Pizza	30940.5
The Four Cheese Pizza	32265.70000000065
The Mexicana Pizza	26780.75
The Five Cheese Pizza	26066.5

ER Diagram



pizzas
pizza_id TEXT
pizza_type_id TEXT
size TEXT
price DOUBLE
Indexes
fk_pizzas_order_details1_i...
pizza_types
pizza_type_id TEXT
name TEXT
category TEXT
ingredients TEXT
Indexes
fk_pizza_types_pizzas1_i...
orders
order_id INT
date DATE
time TIME
Indexes
PRIMARY
order_details
order_details_id INT
order_id INT
pizza_id TEXT
quantity INT
Indexes
PRIMARY
fk_order_details_orders_i...

Key Insights

📊 Category Analysis

- The Thai Chicken Pizza category recorded the highest quantity sold among all pizza categories.
- Sales were concentrated among a few high-performing categories, suggesting strong customer preferences.
- Category-wise analysis revealed opportunities to promote lower-performing categories through targeted offers.

⏰ Customer Ordering Behavior

- Peak ordering activity occurred during 12-1 PM, indicating the busiest business period.
- Most orders were placed during lunch and dinner hours, reflecting typical restaurant demand patterns.
- Understanding hourly demand can help optimize staffing and inventory management.

Key Insights

💰 Revenue Insights

- The top 3 pizza types generated a substantial portion of total revenue.
- Revenue contribution analysis showed that certain pizzas significantly outperformed others despite having similar menu prices.
- High-revenue pizzas can be prioritized in marketing campaigns and promotional offers.

📈 Sales Trend Analysis

- Cumulative revenue showed a steady upward trend over time, indicating consistent business growth throughout the observed period.
- Daily sales analysis revealed an average of 138 pizzas sold per day.
- Revenue trends can help forecast future demand and support inventory planning.

Key Insights

🏆 Advanced Findings

- Revenue contribution percentages highlighted the most profitable pizza categories.
- Ranking pizzas within each category identified the strongest performers and potential menu stars.
- The analysis demonstrated that a small number of pizza types contributed disproportionately to overall revenue, following a typical Pareto (80/20) sales pattern.

Thank You